



PUBLICATION DASHBOARD

Forprosjekt

JANUARY 20, 2021
UNIVERSITY I AGDER
Grimstad

Eksempel på kapitler i en forprosjektrapport. Kapitlene forteller om innhold og ikke nødvendigvis navnet på kapittelet. Forside er ikke inkludert.

1 Project Overview

<Gi et kort sammendrag av prosjektet, og sett prosjektet i kontekst.>

The Norwegian Higher education institution and research sectors publish their research papers to the national research information system in Norway (Cristin). The University of Agder, located in the southern region of Norway is one of the higher education institutions that report and register research activities to Cristin. Since Cristin is a shared platform, this project aims to develop a parallel publication dashboard that highlights and shows publications at the University of Agder.

The dashboard will display overview of research and publications at the university. The main goal of this project is visualizing research activities at the university, using advanced charts and diagrams. Project related information like the name of the researcher, collaborators, study areas, number of current research and other related data will be represented at the dashboard.

2 Project Objectives

<presenter målene med prosjektet, hvorfor dette prosjektet er viktig/hensiktsmessig å gjennomføre. Når prosjektet er ferdig, hvilke mål er da oppnådd.>

This project thesis has various purposes such as, to demonstrate skills and knowledge in software engineering, to demonstrate ability to apply process methodology in software project management, to apply industry standards, frameworks and tools and build a software from scratch, in this case the task “publication dashboard”.

This project needs our knowledge of technologies we have acquired over the years and in addition to that, we need to have more self-thought knowledge on some technologies we have not implemented in school such as, data visualization. Hence, this is a perfect opportunity for us to show what we can achieve and learn what we haven’t so far in the process.

We have been given a deadline for the project and we assume the project will take approximately five months, which means from the beginning of February until early may. By planning out a good schedule, we will be able to finish working on the project on time.

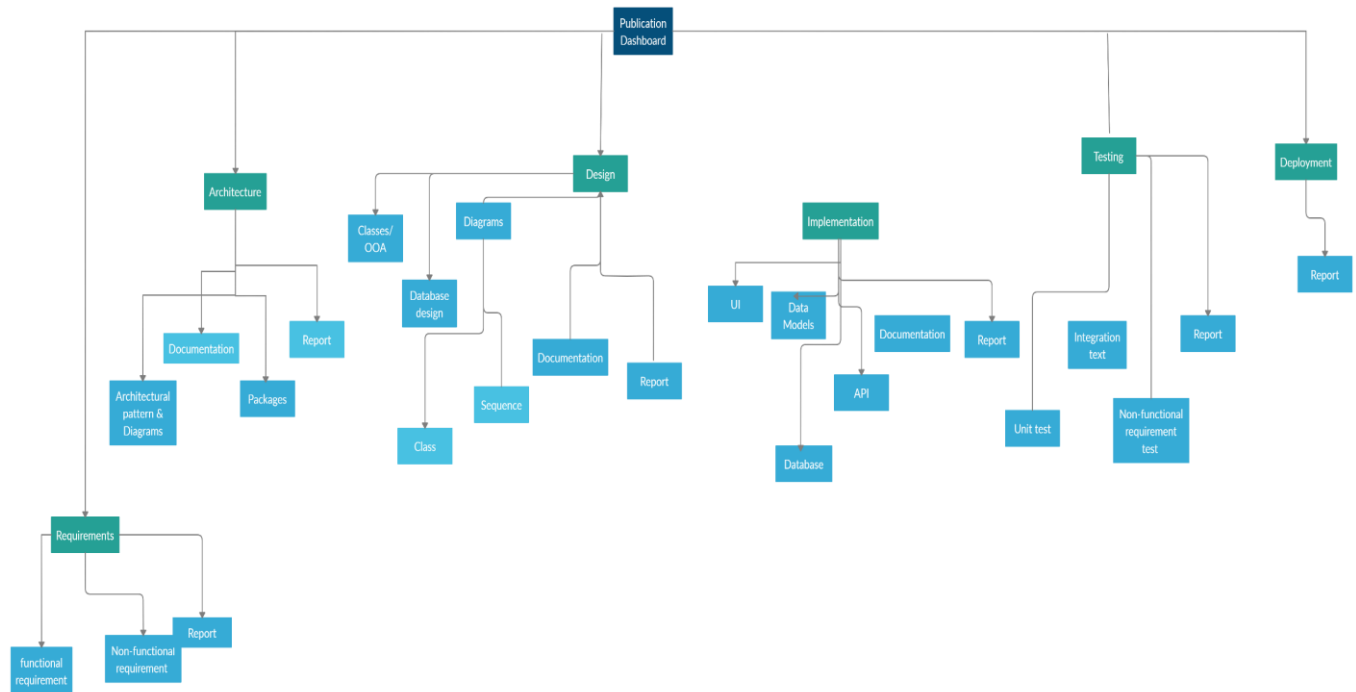
This project focuses mainly on creating the publication dashboard that highlights the research done in UIA. We also achieve good knowledge and experience in various aspects of our field. Hence, what we want to achieve through this task is a successful completion of the project, meaning, the different objectives of the project are met. This furthermore demonstrates the ability of the group to develop a software from inception to production.

3 Project Scope

<I denne delen settes de nå kjente rammene for prosjektet. Her kan forventninger til prosjektet settes, samt avklare grenser for prosjektet.>

3.1 Work Plan

<Den arbeidsmessige planen for prosjektet. Definer hva som må gjøres (ikke hvordan) og stegene som må gjøres for å komme dit. Dette er en logisk fremstilling, ikke en teknisk eller tidsmessig plan. En WBS kan være fint å bruke. >



3.2 In-Scope

<Definer hva som er en del av prosjektet, selv om dette kanskje er innlysende. Elementer som naturlig ikke ville vært en del av prosjektet er det viktig at kommer frem.>

As mentioned in the project overview, the project involves software development. Main features and functionalities of the software is, fetching projects registered by UiA from Cristin, sort and map the data to data-models, stored it in a database (probably temporary storage), analyze and visualize the data using graphs and diagrams. The user interface shall include limited interaction mechanisms, that enable users to choose from a set of limited options and show diagrams that reflect their choice. Registration of research activities is not part of the responsibilities of the system, as reporting and registration is done at the national system. This system, however, shall update its database regularly by frequently requesting and fetching data from Cristin, thus displaying updated information.

3.3 Out-of-Scope

<Definer hva som ikke er den del av prosjektet, men som noen kanskje ville oppfatte som en del av prosjektet. La oss anta at man ville lage en skype klone, men ikke ønsket å ha med skjermdeling bør dette beskrives her. >

Components and capabilities of the product to be built, are discussed earlier. However, confusions and misunderstandings about the product services can easily arise. The project title “Publication Dashboard” may imply different functionalities for whatever reason. To address these and other possible issues, concrete definition of things beyond the project scope follows below.

Services like reporting and registration of research projects are not within the product scope. Whatever data this system utilizes directly comes from Cristin. User authentication and authorization are not needed because, user database is not going to be implemented.

3.3 Deliverables

<Beskriv kort hva som skal leveres. Er det et ferdig produkt, en rapport, kode, etc. >

Elements of the project deliverables include software product, source code, and detailed report. Report comprises thorough discussions and documentations about the project management process, software development methodology, development tools, technologies and frameworks employed during the project period.

4 Project Organization

<Hvis du arbeider med en bedrift kan det være fint å definer roller og oppgaver.>

The team is composed of three competent students. Each member of the team has their own assigned roles. One member takes up the responsibility to lead the team. Main tasks of the team leader are scheduling, directing meetings and preparing agendas for all the meetings. The second student will be responsible for recording everything that happens during the group meetings and discussions. The remaining member manages software development tools like version control software.

For the rest of the tasks and responsibilities, all three members will come together to work in a team to achieve the desired and a remarkable result. Hence, the above organization is a general working structure that guides the team.

5 Schedule

<Her setter man opp Work plan tidsmessig, og gjerne milepælbasert. Dette vil gi dere en kritisk sti for å kunne gjennomføre prosjektet. Det kan være viktig å tenkt enøye gjennom tidsplanen,

spesielt med tanke på (1) goldplating, (2) problemløsning/bugfikning I forhold til rapportskrivning og fremdrift.>

Time Plan and Milestones

As there are 14 weeks left up until the deadline, which is may, 14th, we have planned out our schedule in the following order: -

- Requirement = 1 week
- Architecture = 2 weeks
- Design = 3 weeks
- Implementation = 3 weeks
- Testing = 2 weeks
- Deployment = 1 week
- Editing and other necessary adjustments = 2 weeks

Hence, here is a detailed plan of our project during each week: -

Week 5	Week 6 & 7	Week 8, 9 & 10	Week 11, 12 & 13
-Requirements: finish all the points stated under requirements on the WBS. *Decide: -Programming Language -Frameworks -Tools -Set up development environment -Decide Architecture Pattern -Analysis of the week -Planning for next week	-Architectural pattern and diagram -Packages -Documentation -Report	-Implementable classes -Database designs	-A complete Implementation

Week 14 & 15	Week 16	Week 17, 18 & 19
Testing	Deployment	Editing Final Report Final delivery of project

6 Assumptions and constraints

<Assumptions er uklarte områder i et prosjekt hvor man bare må ta en forutsetning. Dette kan være viktige elementer spesielt i forhold til oppdragsgiver. Eks: Hvis man er usikker på om et stykke software skal kjøre på bare Windows eller både Windows og Linux/Mac, bør man si noe om dette i assumptions, "Vi jobber under forutsetning at softwaren skal kjøre kun på X"
Constraints er en begrensning, zero degree of freedom/wiggle room gitt av oppdragsgiver. F.eks oppdragsgiver vil si at softwaren skal kjøre på Mac, eller at man skal bruke C++. Da har man ikke friheten til å velge noe annet.>

Assumptions about the project

- Software is able to run on all platforms

Constraints about the project

- Use Cristin's API to access and fetch data.

7 Risks

<Risk er usikre aspekter ved prosjektet, dette må monitoreres. Hvis oppdragsgiver skal bistå med utstyr eller lignende, og dette skal bestilles er det en risk hvorvidt dette ankommer til tiden. Hvis flere studenter skal ta opp eksamener i prosjekperioden er dette også en risk for prosjektet. Definer risken, og kom med forslag til skadebegrensning (mitigations).
Husk at risk er produktet av probability og impact.>

Eventually, we might have a few risks during the process. We have gotten a virtual machine from the school, but we might need to rent a server in case we face technical issues in deploying the project to the virtual machine as we once had the same problem during school projects.

Members of the group also might need to retake exams at the same time the project is in process. This might be a risk as it might make us tight on schedule and create a stressful environment.

As a mitigation plan, we have come up with a solution of renting a server and trying to deploy the production in parallel with the virtual machine we received from the client and also getting technical help on many aspects of the project on time.

As for the exams we retake, in the worst-case scenario, it is possible to withdraw from the exams. Other than this, we assume there won't be much of a risk during the production.

8 Critical Success Criteria

<Tenk deg frem til slutten av prosjektet når dere er ferdig og har levert. Hva var det som gjorde at prosjektet ble en suksess? Definer resultatet/funksjonalitetet og ikke deres egen prosess for å nå dit. F.eks «Bedriften kan nå redusere tiden de bruker på å opprette X med Y tid».

Hvis man definerer dette tidlig får man (1) tenkt godt gjennom problemstillingen og (2) har et konkret mål å jobbe mot. Dette kan være viktig mtp bla tidsbruk.>

For the project to be considered a success, there are some criterias it needs to fulfill. Here are some of the criteria: -

- The software is able to fetch correct data from cristin and sort it and temporarily store it. In addition to that, it needs to be able to display statistical diagrams.
- Be able to show the statistics diagrams to the client in the searched manner, meaning if we search by id, then the software is able to show the correct data.
- The university will have its own publication dashboard.
- The time to search and organize researches owned by the University of Agder from Cristin can be reduced by isolating the data locally and showing relevant data when needed.
- Hence, the University is able to use this software for lecture purposes.